

Duy Nguyen

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PROFESSIONAL SUMMARY

Data scientist and MS candidate who designs and runs rigorous ML experiments and builds the data pipelines they run on. Recent research isolated the retrieval encoder, not decoder fine-tuning, as the key lever in medical-imaging RAG; the result is a first-author paper accepted at PSB 2027 with an oral presentation. I own problems end to end: experiment design, modeling, evaluation, and reporting. My replication caught a mislabeled metric in the published benchmark, and my own headline result is validated by a built-in negative control.

EDUCATION

Seattle University

M.S. in Data Science.

Seattle, WA

Expected Jun 2027

UC Berkeley (Online)

Certificate, Machine Learning & AI.

Jul 2024

Simon Fraser University

B.A. in Economics, Data Analysis Concentration.

Burnaby, BC, Canada

May 2023

EXPERIENCE

Graduate Research Assistant, Medical Vision-Language RAG

Spring 2026 – Present

Seattle University · Advisor: Wenjing Yang, PhD

Seattle, WA

- Replicated and audited a published mammography-RAG benchmark on VinDr-Mammo (20,000 images, 5,000 patients) in Python and PyTorch; identified a systematic mislabeling of weighted metrics as macro metrics, correcting the evaluation baseline for all subsequent comparisons.
- Designed and ran a 24-cell controlled experiment (4 encoder substrates x frozen/fine-tuned x 3 vision-language models) with HuggingFace Transformers, ChromaDB, and scikit-learn to isolate the causal driver of generated-report quality.
- Isolated the retrieval encoder as the key performance lever: replacing the frozen OpenCLIP retriever with a domain-native, supervised-contrastively fine-tuned Mammo-CLIP encoder (decoder frozen throughout) raised suspicion macro-F1 from 0.35 to 0.48 (+0.13) and BI-RADS (+0.10) on LLaVA-Med; fine-tuning alone accounts for +0.11 (Qwen2.5-VL) and +0.06 (LLaVA-Med).
- Built the experiment with independent encoder and fine-tuning factors so the causal finding could not rest on a single comparison; carried retrieval-insensitive MedGemma as a negative control to validate the attribution.
- Automated evaluation with reproducible, self-verifying generators (Python/pandas) that recompute 100+ reported metrics byte-exact from raw model predictions; authored a 37-page technical report condensed into a first-author paper accepted at PSB 2027, with an oral presentation (Jan 2027).

Research Data Engineer, Computational Neuroscience Research Group

Spring 2026 – Present

Seattle University · Supervisor: Prof. Brian Fischer

Seattle, WA

- Supported an NIH CRCNS-funded computational neuroscience project on barn owl sound localization by building data infrastructure for neural recordings (full-time over the summer, continuing).
- Engineered an end-to-end ETL pipeline in Python (pandas, SciPy, h5py) converting 5+ proprietary electrophysiology formats (XDPHYS .iid/.itd/.bf, MATLAB .mat, headerless CSV) into a normalized SQLite database of 195 neurons and 1,325 recording passes, with a second researcher's spike-reliability dataset staged and schema-provisioned for load.
- Designed the relational schema and ER model (documented in LaTeX) with foreign-key constraints and explicit data-quality rules covering missing-data handling, tracked exclusions, and path remapping, consolidating scattered lab files into one queryable source of truth.
- Tuned queries and ETL to meet sub-200 ms query and sub-30 s full-load targets, letting lab students self-serve neuron-level queries that previously required manual file handling.
- Ran SQL/SQLite schema and data-integrity audits that caught and corrected issues including all-zero rows and weak-tuning exclusions before downstream analysis.

AWARDS & HONORS

- Winner, Graduate Division, CAUSE Student Data Scrollytelling Contest (2026), Consortium for the Advancement of Undergraduate Statistics Education. Project: "Will AI Make Human Work Worthless or Priceless?": a nine-act interactive data story (D3.js, Scrollama.js) with scroll-driven animation and a reader-controlled parameter widget, translating an original nested-CES general-equilibrium model of AI and cognitive labor into a narrative for non-specialists. Judged on narrative, data interpretation, scrollytelling, and visual design.

SKILLS

ML / AI: PyTorch, HuggingFace Transformers, scikit-learn, Retrieval-Augmented Generation (RAG), vision-language models, QLoRA / parameter-efficient fine-tuning

Experimentation & Causal Inference: experimental design, factorial designs, controls and negative controls, A/B testing, hypothesis testing, effect-size estimation, replication and reproducibility auditing

Data: pandas, NumPy, SciPy, SQLite, ETL design, relational schema / ER design, data quality

Programming: Python, SQL, JavaScript

Tools: Jupyter, Git, LaTeX, D3.js, Scrollama.js, Google Colab (A100 GPU), Ollama